

LABORATORY 5

INTRODUCTION TO 8255A PROGRAMMABLE PERIPHERAL INTERFACE AND EXPERIMENT WITH SEVEN (7)-SEGMENTS DISPLAY AND LED CONNECTION PROGRAM

Objectives:

- To get familiar with 8255A Interface, PIN and port configuration.
- To understand the connectivity between 8255 with I/O ports (P3 and P4), LEDs and 7-segments display units.
- To simulate an example program to display 0-9 digits in 7-segment display units.

Introduction:

8255A is a general purpose programmable I/O device used in microprocessors. It consists of three 8-bit bidirectional I/O ports with 24 I/O pins (figure 1) which may be individually programmed in 2 groups of 12 pins and used in 3 major modes of operation.

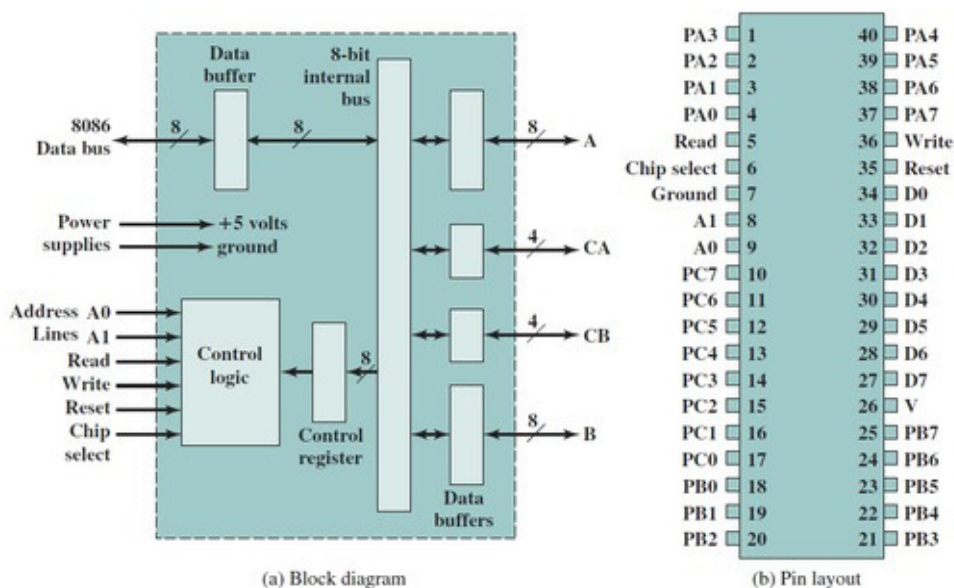


Figure 1: 8255 Functional and PIN OUT Diagram

Carry control and status signal

A1	A0	Function
0	0	Port A
0	1	Port B
1	0	Port C
1	1	Command Register

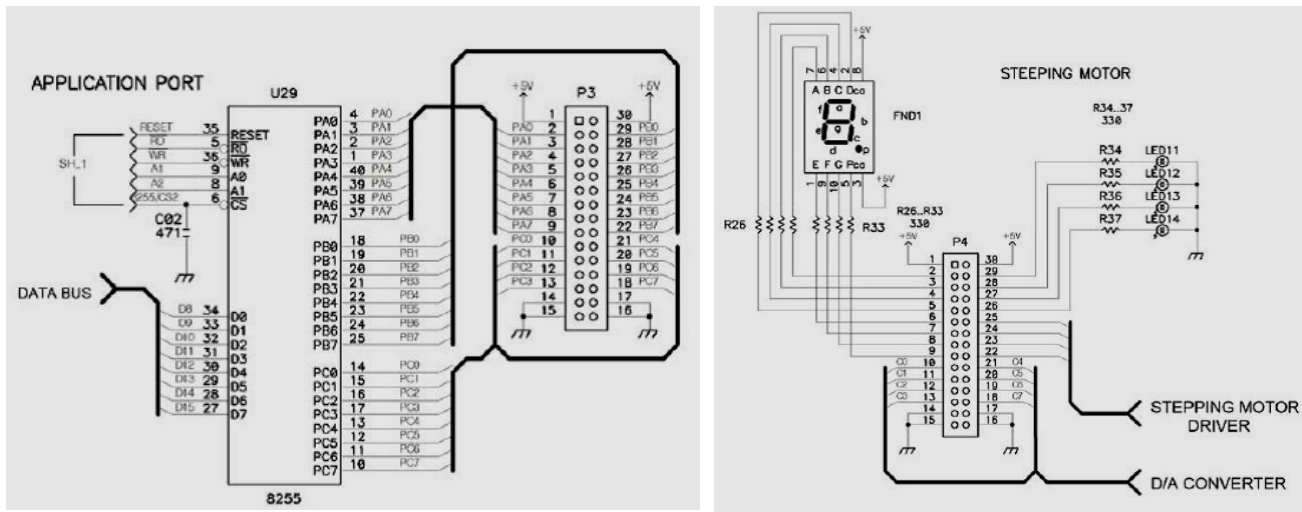


Figure 2: 82C55 interface with Microprocessor and I/O ports (P3, P4)

82C55 has three mode of operation including Mode 0, 1, and 2.

Mode 0- Basic Input/output Mode

Causes 82C55 to function either as a buffered input device or as a latched output device and allows the pins of Group B/Group A to be programmed as simple I/O ports.

Mode 1- Strobe Input/output Mode

Causes operation port A and/or port B to function as latching input devices. Similar to mode 0 but data are transferred through port A/port B and handshaking (DATA READY, ACKNOWLEDGE) and interrupt request signals are provided by port C. Strobe inputs signal to microprocessor retrieve data that are stored into the port registers.

The address of the control register, port A, port B and port C are given below:

```
PPIC_C    EQU 1FH
PPIC      EQU 1DH
PPIB      EQU 1BH
PPIA      EQU 19H
```

Example 1: Display digits 0–9 and some characters on a 7-segment display (Kit mode).

To display the digits 0 – 9 the bit-patterns are given below.

```
P G F E D C B A
1 1 0 0 0 0 0 B
1 1 1 1 1 0 0 1 B
1 0 1 0 0 1 0 0 B
1 0 1 1 0 0 0 0 B
1 0 0 1 1 0 0 1 B
1 0 0 1 0 0 1 0 B
1 0 0 0 0 0 1 0 B
```

```

1 1 1 1 1 0 0 0 B
1 0 0 0 0 0 0 0 B
1 0 0 1 0 0 0 0 B

```

Task 1: Write and run the following program to display 0-9

```

B0 80      MOV AL, 10000000B
E6 1F      OUT 1F, AL          ;GOES TO CONTROL REG
B0 F0      MOV AL, 11110000B
E6 1B      OUT 1B, AL          ;GOES TO PORT B
B0 00      MOV AL, 00000000B
E6 1D      OUT 1D, AL          ; GOES TO PORT C

```

;DISPLAY STARTS HERE

LEVEL: ;should be 100CH if the offset address of the code is 1000H

```

B0 C0      MOV AL, 11000000B
E6 19      OUT 19, AL          ; '0' GOES TO PORT A
B0 F9      MOV AL, 11111001B
E6 19      OUT 19, AL          ; '1' GOES TO PORT A
B0 A4      MOV AL, 10100100B
E6 19      OUT 19, AL          ; '2' GOES TO PORT A
B0 B0      MOV AL, 10110000B
E6 19      OUT 19, AL          ; '3' GOES TO PORT A
B0 99      MOV AL, 10011001B
E6 19      OUT 19, AL          ; '4' GOES TO PORT A
B0 92      MOV AL, 10010010B
E6 19      OUT 19, AL          ; '5' GOES TO PORT A
B0 82      MOV AL, 10000010B
E6 19      OUT 19, AL          ; '6' GOES TO PORT A
B0 F8      MOV AL, 11111000B
E6 19      OUT 19, AL          ; '7' GOES TO PORT A
B0 80      MOV AL, 10000000B
E6 19      OUT 19, AL          ; '8' GOES TO PORT A
B0 90      MOV AL, 10010000B
E6 19      OUT 19, AL          ; '9' GOES TO PORT A

```

```

E6 0C B0 C0 EB D4          ; GIVE THE ADDRESS OF THE INSTRUCTION
MOV AL, 11000000B
10 00      JMP [LEVEL]
00

```

Assignment 1: Write a program to display the hexadecimal digits.

Assignment 2: Write a program to glow the four LEDs of Figure 2 one by one.

Hints: You need to change command register to get the LED on.

```

B0 0F      MOV AL, 00001111B ; B NEED TO BE WORK AS OUTPUT
E6 1B      OUT 1B, AL          ;GOES TO CONTROL REG

```